BIDISHA DASH

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EDUCATION

Siksha 'O' Anusandhan (Deemed to be University) **Odisha, India**
B.Sc (Hons.) Agriculture | Nov 2021 – May 2025 [Degree Certificate]

- **CGPA:** 8.69/10 (First Division with Distinction)
- **Relevant Coursework:** Crop Production Technology, Crop Improvement, Plant Biotechnology, Breeding & Genetics, Plant Pathology & Nematology, Entomology, Problematic Soils, Soil Fertility, Crop Physiology, Agricultural Microbiology, Agricultural Engineering, Geo-informatics and Precision Farming, Statistical Methods, Agri-informatics, Protected Cultivation

TEST SCORES

TOEFL iBT: 97/120 [Score Report] **Aug 8, 2025**

PUBLICATIONS

Multi-Modal Spatial-Temporal Graph Attention Network and Stacking Ensemble Hybrid (ST-GAT-SEH) Framework for Precision Smart Irrigation Control | Manuscript under review **2026**
Second author

- Hybrid ST-GAT and XGBoost/Random Forest stacking model for multi-depth (10, 30, 60 cm) soil moisture forecasting; achieved $R^2 = 0.984$ and RMSE = 0.012 m^3/m^3 in a 120-day field trial, saving 39.6% and 18.2% irrigation water versus schedule-based and threshold-based control, with 14.1% higher yield.

RESEARCH & PROJECTS

Sugarcane Eye Spot Disease Detection System [View Project] **2026**
AWaDH-Annam AI, Indian Institute of Technology Ropar

- Developed a two-stage deep learning pipeline using EfficientNet-B3 for healthy–unhealthy binary classification (99.86% accuracy, 99.91% best validation F1), followed by YOLO11m segmentation for precise disease localization, achieving 84.59% bounding-box precision and 86.19% mAP@50 on a 768×768 validation set.
- Built a novel field-collected dataset of eye spot (*Bipolaris sacchari*) combining farm samples with curated online datasets, and manually annotated 16,506 unhealthy leaf images for instance segmentation.
- Integrated a recommendation engine and confidence-analysis module delivering disease localization, severity ranking, environmental risk conditions, and management practice recommendations.

EXPERIENCE

Agriculture Intern **Jul 13, 2026 – Present**
ANNAM.AI, Indian Institute of Technology Ropar | Punjab, India

- Contributing to agricultural data collection, cleaning, sorting, model testing, and annotation for active AI-for-Agriculture projects, including AVNI: Grow with Nature (plant-scanning educational model for children), FasalNetra (crop pest and disease detection system for Indian farmers), and Sanket (pest risk advisory system).

Agritech Intern | *AI 4 AgriTech Course + Internship Programme* [Certificate] **May 18 – Jul 1, 2026**
ANNAM.AI | *IIT Ropar-TIF (iHub-AWaDH)* | Punjab, India

- Built end-to-end AI pipelines using U-Net, YOLOv5, SVM, ANN, and K-Means for yield prediction, disease detection, and agricultural land zoning, integrating SHAP and LIME for model interpretability.
- Developed an AI-powered sugarcane eye spot disease detection system.

Summer Bioinformatics Internship [Certificate] **Mar 1 – Apr 30, 2026**
Genome Guide | Virtual Internship

- Practical training in Python and R for biological data analysis, Cytoscape (network biology), AutoDock (molecular docking), SRA Toolkit (NGS data), and PyMOL (protein structure).

Technical Trainee **Jul 1 – Dec 31, 2025**
MCKA Agro Consultancy Services | West Bengal, India

- Handled polyhouse cultivation and pro-tray seedling production of high-value crops.

- Scouted over 40 farmers' fields to diagnose crop and soil problems and assisted senior consultants in developing adoptable solutions.
- Assisted senior consultants in conducting training programs.

Experiential Learning Programme **Dec 27, 2024 – Apr 2025**

Siksha 'O' Anusandhan (Deemed to be University) | Odisha, India

- **EL-422: Seed Production Technology –** Produced foundation maize seeds (variety CAL-1415) with proper field isolation, roguing, and clean harvesting.
- **EL-434: Seed Quality Testing & Quality Enhancement –** Conducted seed quality testing (germination, purity, vigor, and moisture) and trialed various seed priming and other quality enhancement treatments.

Student Intern [Certificate] **Sep 2 – Dec 2, 2024**

Rathindra Krishi Vigyan Kendra, Visva-Bharati | West Bengal, India

- Analyzed farmer-submitted soil samples for 12 key parameters (pH, EC, OC, macro- and micronutrients) to issue Soil Health Cards (SHC).
- Supported field research on nano-fertilizer doses in rice; produced organic green gram and prepared biostimulants, natural fungicides-pesticides, vermicompost, and azolla for improved soil and plant health.
- Surveyed over 80 farmers to promote pulse production.

Village Attachment Program **Jul 4 – Aug 31, 2024**

Siksha 'O' Anusandhan (Deemed to be University) | Odisha, India

- Used Participatory Rural Appraisal tools to study the agro-socioeconomic condition of the village.
- Conducted 2 training programs on Integrated Pest Management and paddy straw mushroom, and an awareness program educating farming families on government schemes and subsidies.
- Prepared alternative farm plans for small and marginal farmers to increase income by integrating practices such as mushroom cultivation and biofloc fish farming.

SKILLS

Technical Skills

Machine / Deep Learning: CNN, U-Net, YOLOv5/YOLO11, EfficientNet, ANN, SVM, KNN, K-Means, DBSCAN, Isolation Forest, Transformers; SHAP/LIME

Programming & Libraries: Python, R, SQL; PyTorch, TensorFlow/Keras, Scikit-learn, OpenCV, Albumentations, NumPy, Pandas, SciPy, Matplotlib, Seaborn

Geospatial & Bioinformatics: Google Earth Engine; Cytoscape, AutoDock, PyMOL, SRA Toolkit

Tools: Jupyter, Google Colab, Flask/FastAPI, TFLite; image annotation

Techniques: Data augmentation, hyperparameter tuning (grid search, Bayesian), image segmentation, object detection, anomaly detection

Laboratory & Field Skills

Agronomy & Weed Science: Crop establishment, input calculations, weed management, yield estimation, irrigation scheduling, soil-water measurements, compost and biofertilizer preparation.

Horticulture & Post-harvest: Nursery management, sexual/asexual propagation, orchard layout, training and pruning, polyhouse cultivation, pro-tray seedlings, preservation and quality evaluation.

Plant Breeding & Genetics: Emasculation, hybridization and selfing; pedigree; estimation of heritability and heterosis; RBD field layout and analysis; germplasm and donor-parent characterization

Plant Pathology: Pathogen isolation on media and Koch's postulates; fungal and bacterial staining and identification; field disease diagnosis; herbarium of 50 mounted diseased specimens; mass multiplication of Trichoderma and Pseudomonas; nematode extraction and staining

Entomology & Pest Management: Insect identification and preservation; damage and loss assessment; population sampling; stored-grain pest assessment; mass multiplication of Trichogramma and NPV; IPM planning; beekeeping

Soil, Water & Salinity: Soil pH, EC, CEC and organic carbon; colorimetry and flame photometry for N, P, K, S and Zn; irrigation water quality (EC, pH, carbonate, Ca, Mg); reclamation of saline, sodic and acid soils

Plant Physiology & Microbiology: Photosynthesis and transpiration measurement with IRGA; relative water content; isolation of Rhizobium

TRAINING & CERTIFICATIONS

MOOC on Artificial Intelligence in Agriculture | ICAR–National Academy of Agricultural Research Management (NAARM), Hyderabad | Online
End-to-End Google Earth Engine | Spatial Thoughts | Online
Python for Data Science and Machine Learning Bootcamp | Udemy | Online [Certificate]
R Programming – R Language for Absolute Beginners | Udemy | Online [Certificate]
SQL for Data Science | Great Learning | Online [Certificate]
Practical Geospatial Geostatistical Modeling | Coursera, Case Western Reserve University | Online [Certificate]

PRESENTATIONS

Sugarcane Eye Spot Disease Detection System | AWaDH-Annam AI, Indian Institute of Technology Ropar **2026**
Nutrient, Processing and Value Addition in Millets | National Millet Expo, National Rice Research Institute **2023**

OUTREACH ACTIVITIES

Swachh Bharat Mission **2022 – 2024**
Save Soil Movement **2022**